

Sen Zhang

Registered Architect in NY · zsenarchitect@gmail.com

Summary

Registered NY architect (NCARB), 9 years post-Master's, currently Computational BIM Developer at Ennead Architects. Solo developer behind Ennead's Design Technology stack: EnneadTab (500+ tools, three U.S. Copyright registrations, 60 daily users, roughly half the design staff), plus BimRunner, RenderPolisher, FieldDay, AIA-Sustainable-Reporting, and 6 other production services. Outside Ennead: Realm (real-estate networking platform) and several small-business ventures with AI agents handling daily operations. Speaker: Advancing Computational Design 2026, Autodesk University 2024, Digital Built Week Americas 2023.

Experience

Ennead Architects LLP / Computational BIM Developer

New York, NY · 2022–Present

Architectural Designer, BIM Specialist

2020–2022

- **Keystone developer of Ennead's Design Technology practice and sole owner of the firm's plugin and SaaS stack. Drove voluntary firm-wide adoption to 60 daily users (~50% of design staff) without a top-down mandate or training rollout. Set the firm's on-prem AI/data architecture and authored the partner-approved capital-equipment plan for the 176-machine fleet. Continues to perform BIM manager duties on active projects.**
 - + Developer of EnneadTab and 8+ other firm-internal tools (see Selected Works below)
 - + BIM management on active projects:
 - Healthcare (NE US): NYU Langone, Northwell, Lenox Hill
 - Commercial (Asia): FTZ Shanghai, ByteDance Houhai HQ, Bilibili HQ
 - + Solution architect for large-model BIM data, complex families; in-house training and mentoring
 - + Rhino-to-Revit interoperability via zero-learning-curve internal tooling; triage of complex parametric and industrial-scale geometry

Arquitectonica / Designer, BIM Specialist, CG Visualization

New York, NY · 2017–2020

- 40 Hudson Yards (NYC office tower); Amazon HQ2 master planning (Chicago); 31 Parliament mid-rise residential tower (Toronto SmartCity)
- Curtis Block (Calgary); Warsaw Hotel (Poland); Newport Waterfront; Equinox Hotel (San Jose); Queens Sherbourne mixed-use with historical preservation (Toronto)

RPI Publication Dept. / Book-design, Asset Management

Troy, NY · 2013–2016

Publicity

Advancing Computational Design Conference / Speaker

Austin, TX · 2026

The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency

Autodesk University / Session Speaker

San Diego, CA · 2024

Revit As A Game Engine: Running Board Game Entirely Inside Revit

Digital Built Week Americas / Panel Speaker

Dallas, TX · 2023

Promoting Computational Design to Non-Programmers

Skills & Credentials

- Registered Architect, New York (NCARB)
- Languages: Python (CPython, IronPython), primary; TypeScript / React / Next.js, applied (LLM-assisted)
- Architect API: Revit API, RhinoCommon API, Grasshopper, Dynamo

- Web stack used in production: Next.js 14–16, NextAuth v5, Drizzle ORM, Vercel Postgres / Blob / KV, Resend, Chart.js, react-pdf, GitHub Actions, FastAPI
- AI integration: on-prem Ollama (private deployment, data stays inside firm network); cloud Claude / OpenAI / Gemini; Claude Code agents in production; model-agnostic via MCP
- Spoken: English, Mandarin Chinese (native)

Education

Columbia University GSAPP / M.S. Advanced Architecture Design	New York, NY · 2016–2017
Rensselaer Polytechnic Institute / Bachelor of Architecture	Troy, NY · 2012–2016
Tongji University / Major in Urban Planning	Shanghai · 2010–2012

Awards & Recognition

• EnneadTab, 3 U.S. Copyright registrations, Sen Zhang as sole author	2024
• AEC Hackathon (NYC), Educational Tool Connecting Junior and Senior Students, Team Honorable Mention	2023
• Hyperloop Campus, Las Vegas, Honorable Mention	2020
• Greenway to Ideal City, Chengdu, 2nd Place	2019
• Tokyo Anti-Library, Honorable Mention	2018
• Non Architecture: Dance, Finalist	2018
• AIA Henry Adams Award	2016
• Magna Cum Laude Honors, Dean's List 2012–2016	2016
• George T. Droste Scholarship, Structural engineering study	2015
• Faculty Selection Award, Academic proficiency in Architecture	2015

SELECTED WORKS

Note: the EnneadTab product family and the internal Ennead-owned tools below are Ennead IP and cannot be included here with images. A live walkthrough with sensitive information redacted is available during interview.

Office Tools at Ennead Architects

EnneadTab / Developer

2021–Present

Firm-wide plugin ecosystem across Revit, Rhino, CAD, Adobe, Excel, and Email. Built and maintained solo. Designed for zero-friction adoption: background auto-updates, instant deployment, and an intuitive UX. Spread organically across the firm by user demand. 500+ tools, three U.S. Copyright registrations (2024), 60 active daily users, 17×/person/workday, 63,627 logged events Q4 2025. Stack: Python (CPython + IronPython) + Revit/Rhino APIs.

EnneadTab Self-Healing Agent Ecosystem / Developer

2025–Present

Autonomous self-healing layer wrapping the EnneadTab-OS codebase. Detects production errors across 60 daily users, drafts fixes, and opens pull requests ready for human review and merge. Featured in the Austin 2026 talk on agile tool introduction.

EnneadTabLLM / Developer

2025–2026

On-prem AI assistant for plain-English queries across firm standards, templates, and project history. Client data never leaves the firm network. Featured in the 2026 Austin talk on agile tool introduction.

BimRunner / Developer

2026–Present

Production analytics platform that closes the feedback loop between live BIM data and project authoring. Local harvester ingests 400+ models weekly; web dashboard surfaces historical scoring, trending data, project pin-up wall, scheduled cron reports, and an AI agent that generates custom analysis scripts on demand to answer complex cross-project queries. Used by project leads across the firm to triage live work before coordination crises.

NYU-HQ / Developer

2025–Present

Programming-vs-as-modeled compliance dashboard for the NYU healthcare project. Compares the client area program (Excel) against the live Revit model on a recurring schedule and surfaces gaps as actionable corrections back to the design team. Closes the feedback loop between brief and as-built BIM.

AIA-Sustainable-Reporting / Developer

2026–Present

CRON Excel pipeline + web dashboard for AIA 2030 sustainable-design metrics, with location-based filtering and project-to-project comparison. Replaces the spreadsheet that used to circulate quarterly for AIA 2030 reporting.

RenderPolisher / Developer

2025–2026

AI-driven render post-processing pipeline with deeply customized features tailored to architects' language and delivery workflow. Cleans and uplifts architectural visualizations through a configurable model chain; integrates into the Ennead delivery workflow via the EnneadTab toolbar.

FieldDay / Developer

2026–Present

Offline-capable web app for site inspection on phones, tablets, and desktops. PDF-based spatial pinning, photos, voice notes, and comments with zero-connectivity capture and cloud sync once back online. Preserves spatial accuracy across drawing revisions.

EnneadTabWiki / Developer

2024–Present

Public documentation portal for the EnneadTab tool family. Tool catalog, usage data, and reference content auto-update from the EnneadTab-OS source repository on every release. Docs stay in lockstep with shipped tools, without manual rewrite. Includes an AI-assisted RAG system for semantic search across all tool docs and usage patterns.

EnneadTabHome / Developer

2025–2026

Landing page and secure auth gateway for the EnneadTab application family. Single sign-on via Microsoft 365 / Entra ID, fronting all in-house web services.

EmployeeData / Developer

2025–2026

Decision-support system for Ennead partners on capital-equipment prioritization. Consolidated weekly directory scrapes and per-machine hardware reports into a single employee-by-machine view; surfaced that 70% of the 176-machine fleet was Retired and 81% of GPUs were 5+ years old, sequenced into a quarterly replacement plan through 2027 Q3. Replaced an ad-hoc Excel email workflow; gated to the partners and DT lead who approve capex. Delivered and retired April 2026 once the decision cycle completed.

2025 Performance Review / Developer

2025–2026

Firm-wide employee-evaluation system used by HR to streamline the annual review process. Reads Excel data and generates an interactive HTML report with software-proficiency tracking and visual analytics. Deployed to GitHub Pages.

EnneaDuck / Designer

2022–Present

Community-mascot system across the EnneadTab ecosystem: a pixel-art duck embedded throughout the toolset (achievements, status pings, gamified adoption hooks) plus a Next.js Duckland community page. Per user feedback, made the toolset feel more memorable and approachable, adding the warmth layer that helped firm-wide adoption stick as a voluntary community rather than a top-down rollout.

Independent

Realm / Digital Strategy Architect

2025–Present

Networking platform connecting architects, real-estate operators, and exhibition partners through curated offline events. Multi-step RSVP, admin/client dashboards, role-based partner gating, branded email flows.

Atelier-T / Founder

2025–Present

Early-stage venture for high-end salons. Operating model pairs a customer-facing web app with an agent-team that runs daily small-business operations end-to-end.

Social-Media Ads Service / Lead Engineer

Ongoing

AI-heavy content creation and social-media engagement automation, sold as an ad service to small-business clients.

Social-Media Channel: Android App / Lead Engineer

Ongoing

Native Android consumer app focused on intentional social-media consumption. Early-stage / pre-launch.

PullRequest-For-Revit / Developer

2025

Built at the AECtech NYC 2025 hackathon. Brings a Git-style pull-request review workflow into Revit, letting a project architect gate changes to key model elements before the rest of the team can sync them to the central model. Records element geometry and metadata to Parquet, diffs the snapshot against the live model, and holds flagged changes for peer approval. C# Revit add-in plus a React/WebView2 frontend with an in-Revit DirectContext3D diff overlay.